Chapter 3. Meromorphic Functions and the Logarithm

Zeros and poles, residues, singularities, the argument principle, homotopies, the complex logarithm

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Overview

A general principle, implicit in Riemann's work: analytic functions are essentially characterized by their singularities—globally, holomorphic functions are "effectively" determined by their zeros, and meromorphic functions by their zeros and poles. We study isolated singularities, in order of increasing severity: *removable singularities*, *poles*, and *essential singularities*. Poles lead to the *calculus of residues*, whose key ingredient is $\int_C dz/z = 2\pi i$. Indefinite integrals of functions with singularities, like $\int dz/z = \log z$, may fail to be single-valued; exploiting this gives the *argument principle*, which counts zeros inside a curve and yields the open mapping theorem and the maximum principle. To define the logarithm properly we introduce homotopies and simply connected domains.

1 Zeros and poles

A **point singularity** (or **isolated singularity**) of f is a point z_0 such that f is defined in a neighborhood of z_0 but not at z_0. For $f(z) = z$ on the punctured plane, 0 is removable (set $f(0) = 0$). $g(z) = 1/z$ cannot be extended continuously to 0; it grows to infinity, and 0 is a pole. $h(z) = e^{1/z}$ grows indefinitely as $z \to 0$ along the positive reals, tends to 0 along the negative reals, and oscillates rapidly but stays bounded along the imaginary axis: removable singularities and poles do not tell the whole story.

z_0 is a **zero** of f if $f(z_0) = 0$. By analytic continuation, the zeros of a non-trivial holomorphic function are isolated.

Theorem 1.1

Suppose that f is holomorphic in a connected open set Ω, has a zero at a point $z_0 \in \Omega$, and does not vanish identically in Ω. Then there exists a neighborhood $U \subset \Omega$ of z_0, a non-vanishing holomorphic function g on U, and a unique positive integer n such that $f(z) = (z - z_0)^n g(z)$ for all $z \in U$.

Proof. Since Ω is connected and $f \not\equiv 0$, f is not identically zero near z_0 (Theorem 4.8). Near z_0, $f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k$; let n be the smallest integer with $a_n \neq 0$. Then $f(z) = (z - z_0)^n [a_n + a_{n+1}(z - z_0) + \cdots] = (z - z_0)^n g(z)$, with g holomorphic and non-vanishing near z_0 (as $a_n \neq 0$).

Uniqueness: if also $f(z) = (z - z_0)^m h(z)$ with $h(z_0) \neq 0$ and $m > n$, then $g(z) = (z - z_0)^{m-n} h(z)$ and letting $z \to z_0$ gives $g(z_0) = 0$, a contradiction; $m < n$ similarly forces $h(z_0) = 0$. So $m = n$ and $h = g$. ∎

f has a **zero of order** n (multiplicity n) at z_0; a zero of order 1 is **simple**. A **deleted neighborhood** of z_0 is $\{z : 0 < |z - z_0| < r\}$. A function f defined in a deleted neighborhood of z_0 has a **pole** at z_0 if $1/f$, defined to be zero at z_0, is holomorphic in a full neighborhood of z_0.

Theorem 1.2

If f has a pole at $z_0 \in \Omega$, then in a neighborhood of that point there exist a non-vanishing holomorphic function h and a unique positive integer n such that $f(z) = (z - z_0)^{-n} h(z)$.

Proof. By Theorem 1.1, $1/f(z) = (z - z_0)^n g(z)$ with g holomorphic and non-vanishing near z_0; take $h = 1/g$. ∎

n is the **order** of the pole (it describes the rate of growth); a pole of order 1 is **simple**.

Theorem 1.3

If f has a pole of order n at z_0, then

$$f(z) = \frac{a_{-n}}{(z-z_0)^n} + \frac{a_{-n+1}}{(z-z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z-z_0} + G(z), \tag{1}$$

where G is a holomorphic function in a neighborhood of z_0.

Proof. Expand $h(z) = A_0 + A_1(z-z_0) + \cdots$ in Theorem 1.2; then $f(z) = (z-z_0)^{-n}(A_0 + A_1(z-z_0) + \cdots)$ has the stated form. ∎

The sum $\frac{a_{-n}}{(z-z_0)^n} + \cdots + \frac{a_{-1}}{z-z_0}$ is the **principal part** of f at z_0, and a_{-1} the **residue**, $\operatorname{res}_{z_0} f = a_{-1}$. All other terms of the principal part have primitives in a deleted neighborhood of z_0, so for the principal part P and any circle C centered at z_0, $\frac{1}{2\pi i}\int_C P(z)\,dz = a_{-1}$. For a simple pole, $\operatorname{res}_{z_0} f = \lim_{z\to z_0}(z-z_0)f(z)$.

Theorem 1.4

If f has a pole of order n at z_0, then

$$\operatorname{res}_{z_0} f = \lim_{z\to z_0} \frac{1}{(n-1)!}\Big(\frac{d}{dz}\Big)^{n-1}(z-z_0)^n f(z).$$

Proof. By (1), $(z-z_0)^n f(z) = a_{-n} + a_{-n+1}(z-z_0) + \cdots + a_{-1}(z-z_0)^{n-1} + G(z)(z-z_0)^n$; differentiate $n-1$ times and let $z \to z_0$. ∎

2 The residue formula

Motivation. If f has isolated singularities inside a contour, Cauchy's theorem no longer gives 0; instead each singularity contributes a residue, the coefficient of $(z-z_0)^{-1}$ in the Laurent expansion. This turns the computation of difficult real integrals into finding a few coefficients.

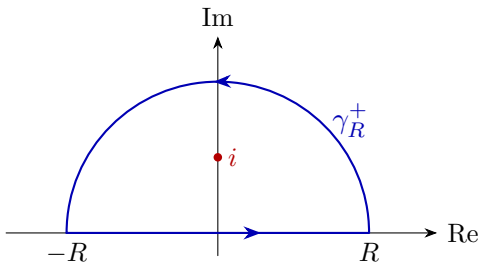


Figure. *The standard semicircular contour for $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$: only the pole at i lies inside, the arc contributes $O(1/R) \to 0$, and the residue formula gives $2\pi i \cdot \frac{1}{2i} = \pi$.*

Theorem 2.1

Suppose that f is holomorphic in an open set containing a circle C and its interior, except for a pole at z_0 inside C. Then

$$\int_C f(z)\,dz = 2\pi i \operatorname{res}_{z_0} f.$$

Proof. Using a keyhole contour avoiding the pole and letting the corridor width tend to 0, $\int_C f = \int_{C_\epsilon} f$, C_ϵ the small circle of radius ϵ about z_0. By the Cauchy integral formula applied to the constant

a_{-1}, $\frac{1}{2\pi i}\int_{C_\epsilon}\frac{a_{-1}}{z-z_0}\,dz = a_{-1}$; by the formulas for derivatives (Corollary 4.2), $\frac{1}{2\pi i}\int_{C_\epsilon}\frac{a_{-k}}{(z-z_0)^k}\,dz = 0$ for $k > 1$; and $\int_{C_\epsilon} G = 0$ by Cauchy's theorem. By (1), $\int_{C_\epsilon} f = 2\pi i a_{-1}$. ■

Corollary 2.2

Suppose that f is holomorphic in an open set containing a circle C and its interior, except for poles at the points $z_1, \ldots, z_N$ inside C. Then

$$\int_C f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

Proof. Use a multiple keyhole with a loop around each pole; letting the corridor widths go to zero, the integral over C becomes a sum of integrals over small circles, to which Theorem 2.1 applies. ■

Corollary 2.3

Suppose that f is holomorphic in an open set containing a toy contour γ and its interior, except for poles at the points $z_1, \ldots, z_N$ inside γ. Then

$$\int_\gamma f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

Here γ has positive orientation.

(Choose a keyhole appropriate to the toy contour and reduce to small circles around the poles.) This identity is the **residue formula**.

2.1 Examples

To compute $\int_{-\infty}^{\infty} f(x)\,dx$, extend f to $\mathbb{C}$ and choose toy contours γ_R with $\lim_{R\to\infty}\int_{\gamma_R} f(z)\,dz = \int_{-\infty}^{\infty} f(x)\,dx$; the residues give $\int_{\gamma_R} f$. Choosing γ_R is the challenge, often guided by the decay of f.

Example 1

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi. \tag{2}$$

(Changing variables $x \mapsto x/y$, this says $\int P_y(x)\,dx = 1$ for the Poisson kernel $P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$.) $f(z) = 1/(1+z^2) = 1/((z-i)(z+i))$ has simple poles at $\pm i$, and $\operatorname{res}_i f = 1/2i$. Let γ_R be the segment $[-R, R]$ together with the upper half-circle C_R^+ of radius R. For large R, $\int_{\gamma_R} f = 2\pi i \cdot \frac{1}{2i} = \pi$. Since $|f(z)| \le B/|z|^2$ on C_R^+, $\left|\int_{C_R^+} f\right| \le \frac{B}{R^2}\pi R \to 0$. Hence $\int_{-\infty}^{\infty} f(x)\,dx = \pi$. (The lower semicircle, with the other pole, works equally well.)

Example 2

An integral that plays an important role in Chapter 6:

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x}\,dx = \frac{\pi}{\sin \pi a}, \qquad 0 < a < 1.$$

Let $f(z) = e^{az}/(1+e^z)$ and γ_R the rectangle with sides on $\mathbb{R}$ and on $\mathrm{Im}(z) = 2\pi$, vertices $\pm R$, $\pm R + 2\pi i$. The only zero of the denominator inside is $z = \pi i$. There

$$(z-\pi i)f(z) = e^{az}\frac{z-\pi i}{e^z - e^{\pi i}}, \qquad \lim_{z\to\pi i}\frac{e^z - e^{\pi i}}{z-\pi i} = e^{\pi i} = -1,$$

so f has a simple pole at πi with residue $-e^{a\pi i}$, and

$$\int_{\gamma_R} f = -2\pi i e^{a\pi i}. \tag{3}$$

Let $I_R = \int_{-R}^{R} f(x)\,dx \to I$. The top side (right to left) contributes $-e^{2\pi i a} I_R$. On the right vertical side $A_R = \{R + it : 0 \le t \le 2\pi\}$, $\left|\int_{A_R} f\right| \le \int_0^{2\pi} \left|\frac{e^{a(R+it)}}{1+e^{R+it}}\right| dt \le Ce^{(a-1)R} \to 0$ since $a < 1$; the left side is bounded by $Ce^{-aR} \to 0$ since $a > 0$. So (3) gives $I - e^{2\pi i a} I = -2\pi i e^{a\pi i}$, whence

$$I = -2\pi i \frac{e^{a\pi i}}{1 - e^{2\pi i a}} = \frac{2\pi i}{e^{\pi i a} - e^{-\pi i a}} = \frac{\pi}{\sin \pi a}.$$

Example 3

$1/\cosh \pi x$ is its own Fourier transform: with $\cosh z = \frac{e^z + e^{-z}}{2}$,

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x\xi}}{\cosh \pi x}\,dx = \frac{1}{\cosh \pi \xi}.$$

Let $f(z) = e^{-2\pi i z\xi}/\cosh \pi z$ and γ_R the rectangle with vertices $\pm R$, $\pm R + 2i$. The denominator vanishes when $e^{2\pi z} = -1$; inside the rectangle the poles are $\alpha = i/2$ and $\beta = 3i/2$. Since

$$(z-\alpha)f(z) = 2e^{-2\pi i z\xi} e^{\pi z}\frac{z-\alpha}{e^{2\pi z} - e^{2\pi\alpha}}, \qquad \lim_{z\to\alpha}(z-\alpha)f(z) = 2e^{-2\pi i\alpha\xi}e^{\pi\alpha}\frac{1}{2\pi e^{2\pi\alpha}} = \frac{e^{\pi\xi}}{\pi i},$$

f has a simple pole at α with residue $e^{\pi\xi}/(\pi i)$; similarly at β with residue $-e^{3\pi\xi}/(\pi i)$. On $z = R + iy$, $0 \le y \le 2$: $|e^{-2\pi i z\xi}| \le e^{4\pi|\xi|}$ and $|\cosh \pi z| \ge \frac{1}{2}(e^{\pi R} - e^{-\pi R}) \to \infty$, so the vertical sides contribute 0 in the limit. Since $\cosh \pi z$ is periodic with period $2i$, the top side contributes $-e^{4\pi\xi} I$. The residue formula gives

$$I - e^{4\pi\xi} I = 2\pi i\Big(\frac{e^{\pi\xi}}{\pi i} - \frac{e^{3\pi\xi}}{\pi i}\Big) = -2e^{2\pi\xi}(e^{\pi\xi} - e^{-\pi\xi}),$$

and since $1 - e^{4\pi\xi} = -e^{2\pi\xi}(e^{2\pi\xi} - e^{-2\pi\xi})$,

$$I = 2\frac{e^{\pi\xi} - e^{-\pi\xi}}{e^{2\pi\xi} - e^{-2\pi\xi}} = \frac{2}{e^{\pi\xi} + e^{-\pi\xi}} = \frac{1}{\cosh \pi\xi}.$$

A similar argument gives, for $0 < a < 1$, with $\sinh z = (e^z - e^{-z})/2$,

$$\int_{-\infty}^{\infty} e^{-2\pi i x\xi}\frac{\sin \pi a}{\cosh \pi x + \cos \pi a}\,dx = \frac{2\sinh 2\pi a\xi}{\sinh 2\pi\xi};$$

$a = 1/2$ is the case above. This yields the Poisson kernel for the strip, and is used in the two-squares theorem (Chapter 10).

3 Singularities and meromorphic functions

Motivation. Isolated singularities come in three kinds, removable, poles and essential, and they behave very differently: near an essential singularity f comes arbitrarily close to every value (Casorati–Weierstrass). Classifying them tells us which functions can be treated as holomorphic functions with a few manageable poles.

Let f be holomorphic in Ω except possibly at $z_0 \in \Omega$. If f can be defined at z_0 so that it becomes holomorphic in all of Ω, z_0 is a **removable singularity**.

Theorem 3.1 (Riemann's theorem on removable singularities)

Suppose that f is holomorphic in an open set Ω except possibly at a point z_0 in Ω. If f is bounded on $\Omega - \{z_0\}$, then z_0 is a removable singularity.

Proof. Take a small disc D centered at z_0 with $\overline{D} \subset \Omega$, boundary circle C positively oriented. We show that for $z \in D$, $z \neq z_0$,

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta. \tag{4}$$

By Theorem 5.4 the right side is holomorphic in all of D and agrees with f for $z \neq z_0$, giving the extension. To prove (4), fix z and use a multiple keyhole avoiding z and z_0; letting the corridors shrink,

$$\int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta + \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = 0,$$

where γ_ϵ, γ'_ϵ are small negatively oriented circles of radius ϵ about z and z_0. As in the proof of the Cauchy integral formula, $\int_{\gamma_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta = -2\pi i f(z)$. Since f is bounded and $\zeta \in \gamma'_\epsilon$ stays away from z, $\left| \int_{\gamma'_\epsilon} \frac{f(\zeta)}{\zeta - z}\, d\zeta \right| \leq C\epsilon$. Let $\epsilon \to 0$. ∎

Corollary 3.2

Suppose that f has an isolated singularity at the point z_0. Then z_0 is a pole of f if and only if $|f(z)| \to \infty$ as $z \to z_0$.

Proof. If z_0 is a pole, $1/f$ has a zero at z_0, so $|f(z)| \to \infty$. Conversely, if $|f(z)| \to \infty$, then $1/f$ is bounded near z_0 and $1/|f(z)| \to 0$; so $1/f$ has a removable singularity at z_0 and vanishes there, i.e. z_0 is a pole. ∎

Isolated singularities fall into three categories: **removable** (f bounded near z_0), **poles** ($|f(z)| \to \infty$ as $z \to z_0$), and **essential singularities** (all others), e.g. $e^{1/z}$ at 0. Near an essential singularity a function typically behaves erratically:

Theorem 3.3 (Casorati–Weierstrass)

Suppose f is holomorphic in the punctured disc $D_r(z_0) - \{z_0\}$ and has an essential singularity at z_0. Then, the image of $D_r(z_0) - \{z_0\}$ under f is dense in the complex plane.

Proof. If not, there are $w \in \mathbb{C}$ and $\delta > 0$ with $|f(z) - w| > \delta$ on the punctured disc. Then $g(z) = 1/(f(z) - w)$ is holomorphic on the punctured disc and bounded by $1/\delta$, so it has a removable singularity at z_0 (Theorem 3.1). If $g(z_0) \neq 0$, $f(z) - w = 1/g(z)$ is holomorphic at z_0; if $g(z_0) = 0$, $f(z) - w$ has a pole at z_0. Both contradict that z_0 is essential. ∎

Picard proved much more: under these hypotheses f takes every complex value infinitely many times with at most one exception.

Definition (meromorphic function)

A function f on an open set Ω is **meromorphic** if there is a sequence $\{z_0, z_1, z_2, \dots\}$ with no limit points in Ω such that (i) f is holomorphic in $\Omega - \{z_0, z_1, \dots\}$, and (ii) f has poles at the points $z_0, z_1, \dots$.

If f is holomorphic for all large z, let $F(z) = f(1/z)$, holomorphic in a deleted neighborhood of 0. f has a **pole at infinity** if F has a pole at 0; similarly an **essential singularity at infinity**, or a **removable singularity** (holomorphic) at infinity. A meromorphic function in $\mathbb{C}$ that is holomorphic or has a pole at infinity is **meromorphic in the extended complex plane**. The principle stated at the outset appears here in its simplest form:

Theorem 3.4

The meromorphic functions in the extended complex plane are the rational functions.

Proof. $f(1/z)$ has a pole or removable singularity at 0, hence is holomorphic in a deleted neighborhood of 0; so f has only finitely many poles in the plane, say $z_1, \dots, z_n$. Near each z_k, $f(z) = f_k(z) + g_k(z)$ with f_k the principal part (a polynomial in $1/(z - z_k)$) and g_k holomorphic near z_k. Similarly $f(1/z) = \tilde{f}_\infty(z) + \tilde{g}_\infty(z)$ with $\tilde{g}_\infty$ holomorphic near 0 and $\tilde{f}_\infty$ a polynomial in $1/z$; let $f_\infty(z) = \tilde{f}_\infty(1/z)$, a polynomial. Then $H = f - f_\infty - \sum_{k=1}^n f_k$ is entire (removable singularities at the z_k) and bounded ($H(1/z)$ is bounded near 0, as the principal part at ∞ was subtracted). By Liouville, H is constant, so f is rational. ∎

Consequently, a rational function is determined up to a multiplicative constant by the locations and multiplicities of its zeros and poles.

The Riemann sphere. In $\mathbb{R}^3$ with coordinates (X, Y, Z), identify the XY-plane with $\mathbb{C}$, and let $\mathbb{S}$ be the sphere of radius $1/2$ centered at $(0, 0, 1/2)$, with north pole $N = (0, 0, 1)$. For $W = (X, Y, Z) \in \mathbb{S} - \{N\}$, the line through N and W meets the XY-plane in a single point $w = x + iy$, the **stereographic projection** of W; this is a bijection between $\mathbb{S} - \{N\}$ and $\mathbb{C}$, given by

$$x = \frac{X}{1-Z}, \quad y = \frac{Y}{1-Z}; \qquad X = \frac{x}{x^2+y^2+1}, \quad Y = \frac{y}{x^2+y^2+1}, \quad Z = \frac{x^2+y^2}{x^2+y^2+1}.$$

As $|w| \to \infty$, $W \to N$, so N is the natural "point at infinity". The extended complex plane is thus the sphere $\mathbb{S}$, the **Riemann sphere** (the *one-point compactification* of $\mathbb{C}$). The point at infinity is on equal footing with the others, and a meromorphic function on the extended plane is a map of $\mathbb{S}$ to itself, sending poles to N.

4 The argument principle and applications

$\log f(z)$ is in general multiple-valued, but it must equal $\log|f(z)| + i \arg f(z)$, and its derivative $f'(z)/f(z)$ is single-valued; $\int_\gamma \frac{f'(z)}{f(z)}\,dz$ is the change in the argument of f along γ, determined (for closed γ) by the zeros and poles of f inside γ. While $\log(f_1 f_2) = \log f_1 + \log f_2$ fails in general, additivity holds for logarithmic derivatives:

$$\frac{(f_1 f_2)'}{f_1 f_2} = \frac{f_1'}{f_1} + \frac{f_2'}{f_2}, \qquad \frac{\left(\prod_{k=1}^N f_k\right)'}{\prod_{k=1}^N f_k} = \sum_{k=1}^N \frac{f_k'}{f_k}.$$

If f has a zero of order n at z_0, $f(z) = (z - z_0)^n g(z)$ with g non-vanishing near z_0, so $\frac{f'(z)}{f(z)} = \frac{n}{z-z_0} + G(z)$, $G = g'/g$ holomorphic: f'/f has a simple pole with residue n. If f has a pole of order n, $f(z) = (z - z_0)^{-n} h(z)$ and $\frac{f'(z)}{f(z)} = \frac{-n}{z-z_0} + H(z)$. The residue formula gives:

Theorem 4.1 (Argument principle)

Suppose f is meromorphic in an open set containing a circle C and its interior. If f has no poles and never vanishes on C, then

$$\frac{1}{2\pi i}\int_C \frac{f'(z)}{f(z)}\,dz = (\text{number of zeros of } f \text{ inside } C) - (\text{number of poles of } f \text{ inside } C),$$

where the zeros and poles are counted with their multiplicities.

Corollary 4.2

The above theorem holds for toy contours.

Three applications: Rouché's theorem (a continuity statement), the open mapping theorem, and the maximum modulus principle.

Theorem 4.3 (Rouché's theorem)

Suppose that f and g are holomorphic in an open set containing a circle C and its interior. If $|f(z)| > |g(z)|$ for all $z \in C$, then f and $f+g$ have the same number of zeros inside the circle C.

Proof. For $t \in [0,1]$ let $f_t(z) = f(z) + tg(z)$, so $f_0 = f$, $f_1 = f+g$, and let n_t be the number of zeros of f_t inside C (with multiplicities). $|f| > |g|$ on C implies f_t has no zeros on C, and by the argument principle $n_t = \frac{1}{2\pi i}\int_C \frac{f_t'(z)}{f_t(z)}\,dz$. $f_t'(z)/f_t(z)$ is jointly continuous in $t \in [0,1]$, $z \in C$ (numerator and denominator are, and the denominator does not vanish on C), so n_t is continuous in t. An integer-valued continuous function on $[0,1]$ is constant (intermediate value theorem), so $n_0 = n_1$. ■

A mapping is **open** if it maps open sets to open sets.

Theorem 4.4 (Open mapping theorem)

If f is holomorphic and non-constant in a region Ω, then f is open.

Proof. Let $w_0 = f(z_0)$; we show all w near w_0 are in the image. Write $g(z) = f(z) - w = (f(z) - w_0) + (w_0 - w) = F(z) + G(z)$. Choose $\delta > 0$ with $\{|z - z_0| \le \delta\} \subset \Omega$ and $f(z) \neq w_0$ on $|z - z_0| = \delta$ (zeros of $f - w_0$ are isolated), then $\epsilon > 0$ with $|f(z) - w_0| \ge \epsilon$ on $|z - z_0| = \delta$. If $|w - w_0| < \epsilon$, then $|F| > |G|$ on that circle, and by Rouché's theorem $g = F + G$ has a zero inside, since F has one (z_0). ■

The maximum of f in Ω means the maximum of $|f|$ in Ω.

Theorem 4.5 (Maximum modulus principle)

If f is a non-constant holomorphic function in a region Ω, then f cannot attain a maximum in Ω.

Proof. If f attained a maximum at z_0, then since f is open, the image $f(D)$ of a small disc D centered at z_0 is open and contains $f(z_0)$; so there are $z \in D$ with $|f(z)| > |f(z_0)|$. ■

Corollary 4.6

Suppose that Ω is a region with compact closure $\overline{\Omega}$. If f is holomorphic on Ω and continuous on $\overline{\Omega}$ then

$$\sup_{z\in\Omega}|f(z)| \le \sup_{z\in\overline{\Omega}-\Omega}|f(z)|.$$

Proof. $|f|$ attains its maximum on the compact set $\overline{\Omega}$, and not in Ω if f is non-constant; if f is constant the conclusion is trivial. ■

Remark

Compactness (boundedness) of $\overline{\Omega}$ is essential. Let Ω be the open first quadrant and $F(z) = e^{-iz^2}$. F is entire, $|F| = 1$ on both boundary half-lines $z = x$, $z = iy$, yet F is unbounded in Ω: $F(z) = e^{r^2}$ for $z = r\sqrt{i} = re^{i\pi/4}$.

5 Homotopies and simply connected domains

The key to the general Cauchy theorem and to multiple-valued functions is knowing in what regions a holomorphic function has a primitive—a global question (relevant to the logarithm, the primitive of $1/z$).

Definition (homotopic curves)

Let $\gamma_0, \gamma_1 \subset \Omega$ be curves with parametrizations on $[a, b]$ and common end-points $\gamma_0(a) = \gamma_1(a) = \alpha$, $\gamma_0(b) = \gamma_1(b) = \beta$. They are **homotopic in** Ω if for each $0 \le s \le 1$ there is a curve $\gamma_s \subset \Omega$, parametrized by $\gamma_s(t)$ on $[a, b]$, with $\gamma_s(a) = \alpha$, $\gamma_s(b) = \beta$ for every s, $\gamma_s(t)|_{s=0} = \gamma_0(t)$, $\gamma_s(t)|_{s=1} = \gamma_1(t)$, and $\gamma_s(t)$ jointly continuous in $s \in [0, 1]$, $t \in [a, b]$.

Loosely, one curve can be deformed into the other continuously without leaving Ω.

Theorem 5.1

If f is holomorphic in Ω, then $\int_{\gamma_0} f(z)\,dz = \int_{\gamma_1} f(z)\,dz$ whenever the two curves γ_0 and γ_1 are homotopic in Ω.

Proof. The key is: if two curves are close and have the same end-points, the integrals agree. $F(s,t) = \gamma_s(t)$ is continuous on $[0,1] \times [a,b]$; its image K is compact. There is $\epsilon > 0$ such that every disc of radius 3ϵ centered at a point of K lies in Ω: otherwise for every $\ell \ge 1$ there are $z_\ell \in K$, $w_\ell \in \Omega^c$ with $|z_\ell - w_\ell| < 1/\ell$; a subsequence $z_{\ell_k} \to z \in K$, then $w_{\ell_k} \to z$ and $z \in \Omega^c$ (Ω^c is closed), a contradiction. By uniform continuity of F, choose δ with $\sup_t|\gamma_{s_1}(t) - \gamma_{s_2}(t)| < \epsilon$ whenever $|s_1 - s_2| < \delta$.

Fix such s_1, s_2. Choose discs $D_0, \dots, D_n$ of radius 2ϵ and consecutive points $z_0, \dots, z_{n+1}$ on γ_{s_1}, $w_0, \dots, w_{n+1}$ on γ_{s_2}, such that the discs cover both curves and $z_i, z_{i+1}, w_i, w_{i+1} \in D_i$, with $z_0 = w_0$ the common beginning point and $z_{n+1} = w_{n+1}$ the common end-point. On each D_i let F_i be a primitive of f (Theorem 2.1). On $D_i \cap D_{i+1}$, F_i and F_{i+1} differ by a constant, so $F_{i+1}(z_{i+1}) - F_i(z_{i+1}) = F_{i+1}(w_{i+1}) - F_i(w_{i+1})$, i.e.

$$F_{i+1}(z_{i+1}) - F_{i+1}(w_{i+1}) = F_i(z_{i+1}) - F_i(w_{i+1}). \tag{5}$$

Hence

$$\int_{\gamma_{s_1}} f - \int_{\gamma_{s_2}} f = \sum_{i=0}^{n}[F_i(z_{i+1}) - F_i(z_i)] - \sum_{i=0}^{n}[F_i(w_{i+1}) - F_i(w_i)] = \sum_{i=0}^{n}\left[F_i(z_{i+1}) - F_i(w_{i+1}) - (F_i(z_i) - F_i(w_i))\right],$$

which telescopes by (5) to $F_n(z_{n+1}) - F_n(w_{n+1}) - (F_0(z_0) - F_0(w_0)) = 0$. Subdividing $[0,1]$ into intervals of length $< \delta$, finitely many applications take us from γ_0 to γ_1. ■

Definition (simply connected)

A region Ω is **simply connected** if any two curves in Ω with the same end-points are homotopic.

Example 1

A disc D is simply connected: $\gamma_s(t) = (1-s)\gamma_0(t) + s\gamma_1(t)$ lies on the segment joining $\gamma_0(t)$ and $\gamma_1(t)$, hence in D. The same works for rectangles and any open convex set.

Example 2

The slit plane $\Omega = \mathbb{C} - (-\infty, 0]$ is simply connected: write $\gamma_j(t) = r_j(t)e^{i\theta_j(t)}$ with $r_j > 0$ and θ_j continuous, $|\theta_j(t)| < \pi$, and let $\gamma_s(t) = r_s(t)e^{i\theta_s(t)}$ with $r_s = (1-s)r_0 + sr_1$, $\theta_s = (1-s)\theta_0 + s\theta_1$; then $\gamma_s(t) \in \Omega$.

Example 3

With some effort one shows that the interior of a toy contour is simply connected, by dividing it into several subregions.

Example 4

The punctured plane $\mathbb{C} - \{0\}$ is not simply connected (proved after Corollary 5.3).

Theorem 5.2

Any holomorphic function in a simply connected domain has a primitive.

Proof. Fix $z_0 \in \Omega$ and define $F(z) = \int_\gamma f(w)\,dw$, the integral over any curve $\gamma \subset \Omega$ joining z_0 to z; this is independent of γ by Theorem 5.1, since Ω is simply connected. Then $F(z+h) - F(z) = \int_\eta f(w)\,dw$ with η the segment from z to $z+h$, and as in the proof of Theorem 2.1, $\lim_{h\to 0} \frac{F(z+h)-F(z)}{h} = f(z)$. ■

Corollary 5.3

If f is holomorphic in the simply connected region Ω, then $\int_\gamma f(z)\,dz = 0$ for any closed curve γ in Ω.

That $\mathbb{C} - \{0\}$ is not simply connected now follows rigorously since $\int_C dz/z = 2\pi i \neq 0$ over the unit circle.

6 The complex logarithm

For $z = re^{i\theta}$ it is natural to set $\log z = \log r + i\theta$ ($\log r$ the standard logarithm of $r > 0$), but θ is unique only up to a multiple of 2π. Locally a continuous choice of θ is determined, but not globally:

winding once around the origin changes the logarithm by $2\pi i$. To obtain a single-valued function we must restrict the domain—the choice of a **branch** or **sheet** of the logarithm.

Theorem 6.1

Suppose that Ω is simply connected with $1 \in \Omega$, and $0 \notin \Omega$. Then in Ω there is a branch of the logarithm $F(z) = \log_\Omega(z)$ so that (i) F is holomorphic in Ω, (ii) $e^{F(z)} = z$ for all $z \in \Omega$, (iii) $F(r) = \log r$ whenever r is a real number and near 1.

So each branch $\log_\Omega(z)$ extends the standard logarithm.

Proof. $1/z$ is holomorphic in Ω; define $\log_\Omega(z) = F(z) = \int_\gamma \frac{dw}{w}$, γ any curve in Ω from 1 to z (path-independent as Ω is simply connected). As in Theorem 5.2, F is holomorphic with $F'(z) = 1/z$: (i). For (ii), $\frac{d}{dz}(ze^{-F(z)}) = e^{-F(z)} - zF'(z)e^{-F(z)} = (1 - zF'(z))e^{-F(z)} = 0$, so $ze^{-F(z)}$ is constant on the connected Ω (Corollary 3.4); at $z = 1$, $F(1) = 0$, so the constant is 1. For (iii), for real r near 1 integrate along the real segment: $F(r) = \int_1^r \frac{dx}{x} = \log r$. ∎

In the slit plane $\Omega = \mathbb{C} - (-\infty, 0]$ we have the **principal branch** $\log z = \log r + i\theta$ for $z = re^{i\theta}$, $|\theta| < \pi$: integrate along the segment from 1 to r and the arc from r to z,

$$\log z = \int_1^r \frac{dx}{x} + \int_{\text{arc}} \frac{dw}{w} = \log r + \int_0^\theta \frac{ire^{it}}{re^{it}}\, dt = \log r + i\theta.$$

In general $\log(z_1 z_2) \neq \log z_1 + \log z_2$: for $z_1 = z_2 = e^{2\pi i/3}$, $\log z_1 = \log z_2 = 2\pi i/3$, while $z_1 z_2 = e^{-2\pi i/3}$ and $\log(z_1 z_2) = -2\pi i/3$. For the principal branch,

$$\log(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \cdots = -\sum_{n=1}^{\infty} (-1)^n \frac{z^n}{n} \qquad \text{for } |z| < 1, \tag{6}$$

since both sides have derivative $1/(1+z)$ and vanish at $z = 0$.

For Ω simply connected with $1 \in \Omega$, $0 \notin \Omega$, and $\alpha \in \mathbb{C}$, define $z^\alpha = e^{\alpha \log z}$ (branch with $\log 1 = 0$). Then $1^\alpha = 1$, and for $\alpha = 1/n$, $(z^{1/n})^n = \prod_{k=1}^n e^{\frac{1}{n}\log z} = e^{\log z} = z$.

Theorem 6.2

If f is a nowhere vanishing holomorphic function in a simply connected region Ω, then there exists a holomorphic function g on Ω such that $f(z) = e^{g(z)}$.

g may be denoted $\log f(z)$; it determines a "branch" of that logarithm.

Proof. Fix $z_0 \in \Omega$ and let $g(z) = \int_\gamma \frac{f'(w)}{f(w)}\, dw + c_0$, γ any path in Ω from z_0 to z, with $e^{c_0} = f(z_0)$ (path-independent as Ω is simply connected). As in Theorem 2.1, g is holomorphic with $g' = f'/f$, and $\frac{d}{dz}(f(z)e^{-g(z)}) = 0$, so $f(z)e^{-g(z)}$ is constant; at z_0 it is $f(z_0)e^{-c_0} = 1$. ∎

7 Fourier series and harmonic functions

Motivation. Real and imaginary parts of holomorphic functions are harmonic, and on the unit circle a holomorphic power series becomes a Fourier series. This section uses these links to derive the mean value property for harmonic functions.

There is a simple and direct relation between Fourier series on the circle and power series of holomorphic functions in the disc. Suppose f is holomorphic in a disc $D_R(z_0)$, so $f(z) = \sum_{n=0}^\infty a_n (z - z_0)^n$ there.

Theorem 7.1

The coefficients of the power series expansion of f are given by

$$a_n = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta$$

for all $n \geq 0$ and $0 < r < R$. Moreover,

$$0 = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta$$

whenever $n < 0$.

Proof. Since $f^{(n)}(z_0) = a_n n!$, the Cauchy integral formula over the circle γ of radius r about z_0 gives, with $\zeta = z_0 + re^{i\theta}$,

$$a_n = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{(re^{i\theta})^{n+1}} rie^{i\theta}\, d\theta = \frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta.$$

For $n < 0$ the same computation gives $\frac{1}{2\pi r^n} \int_0^{2\pi} f(z_0 + re^{i\theta}) e^{-in\theta}\, d\theta = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta$, and since $-n > 0$, $f(\zeta)(\zeta - z_0)^{-n-1}$ is holomorphic in the disc; by Cauchy's theorem the integral vanishes. ■

So the Fourier coefficients of the restriction of f to a circle vanish for $n < 0$, and for $n \geq 0$ equal (up to r^n) the power series coefficients—another special characteristic of holomorphic functions. Since $a_0 = f(z_0)$:

Corollary 7.2 (Mean-value property)

If f is holomorphic in a disc $D_R(z_0)$, then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta})\, d\theta \qquad \text{for any } 0 < r < R.$$

Corollary 7.3

If f is holomorphic in a disc $D_R(z_0)$, and $u = \mathrm{Re}(f)$, then

$$u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta})\, d\theta \qquad \text{for any } 0 < r < R.$$

This follows by taking real parts. Every harmonic function u in a disc is the real part of a holomorphic function there, so the mean-value property holds for all harmonic functions in a disc.